

Nilesh Raj

+91-8887522416 | nileshraj1735@gmail.com | linkedin.com/in/nilesh-raj-nr1735

SUMMARY

AI/ML Developer focused on building practical applications with large language models and retrieval pipelines. I have shipped working tools using Python, LangChain, FAISS, and the OpenAI and Claude APIs, covering everything from document ingestion to structured output delivery via REST APIs. I also use n8n to automate repetitive workflows. NVIDIA-certified in LLM application development.

EDUCATION

United University

BCA with IBM Specialization (AI & DS) | CGPA: 7.15/10

Prayagraj, UP

Jun. 2022 - May 2025

PROJECTS

Contract Risk Analyser | Python, LangChain, OpenAI API, FAISS, Pydantic, Streamlit

2025

- Built a document analysis tool that breaks legal contracts into chunks, searches for relevant clauses using FAISS, and generates structured risk summaries through the OpenAI API.
- Used Pydantic to enforce a consistent output format with risk category, severity, and recommendation fields for every clause, so the results were always predictable and usable.
- Connected the whole pipeline to a Streamlit dashboard via REST API so that people without a technical background could review the results without any manual steps.

Student Doubt Resolution Bot | Python, LangChain, OpenAI API, FAISS, Pydantic, Streamlit, n8n

2025

- Built a question-answering tool that reads course PDFs and lecture transcripts into a FAISS vector store and gives students instant, source-cited answers instead of waiting for instructor responses.
- Added a confidence check so the bot says it does not have enough information from the course materials rather than guessing, keeping the answers reliable.
- Logged every interaction with Pydantic, capturing the question, retrieved context, answer, confidence, and source. Used n8n to send instructors a weekly summary of questions the bot could not answer.

EXPERIENCE

AI/ML Trainee (Internship)

Feb. 2026 - May 2026

Technosavvy Education Technology

Prayagraj, UP

- Built a text classifier using BERT and Hugging Face that automatically sorted 500+ student course reviews into sentiment categories, reaching 88% accuracy after iterative tuning.
- Reduced the time instructors spent manually reading reviews by 60% and improved the model accuracy by 15% over the starting baseline.
- Set up a Streamlit dashboard backed by SQLite that ran on a weekly schedule, giving 10+ mentors a live view of student feedback across 5+ courses.

TECHNICAL SKILLS

AI & GenAI: LLMs, RAG Pipelines, Prompt Engineering, NLP, Sentiment Analysis, BERT, Transformers, Semantic Chunking, Embeddings, Vector Search, Machine Learning

Data Engineering & ML: Python, SQL, Scikit-learn, Hugging Face, Pandas, NumPy, FAISS, Vector Databases, Data Preprocessing, Model Evaluation, SQLite

Frameworks & Infrastructure: LangChain (LCEL), OpenAI API, Anthropic Claude API, NVIDIA NIM, Pydantic, REST APIs, Git, Streamlit, Jupyter Notebook

Exposure: AWS (S3, Lambda), n8n (workflow automation), LangGraph, PostgreSQL, Pinecone, ChromaDB

CERTIFICATIONS

Building LLM Applications – NVIDIA Deep Learning Institute

2025

- Built LLM-powered applications using NVIDIA NIM (Llama 3.1) and LangChain with live API integration and Pydantic-structured output validation across a capstone agentic workflow.